

Project 02: Structural Refinement & Validation for Oncology Research

Computational evaluation, refinement, and pre/post comparison of an oncology-related protein structure

PDB Selection → Raw Structure Retrieval → Structure Inspection → Pre-Validation → GalaxyRefine → Post-Validation → Pre/Post Comparison → Structural Visualization → Oncology Interpretation → Final Report

Overview: students select an oncology-related PDB structure, validate it (ERRAT, PROCHECK, Verify3D) before and after refining it with GalaxyRefine, then quantitatively compare quality metrics and interpret the results in an oncology research context.

Step 1 — Select the Protein Structure

- Select the assigned oncology-related structure from the provided PDB list.

Expected Output:

- Table 1 — Basic Information: protein name, PDB ID, organism, cancer/oncology relevance, mutation (if any), experimental method, resolution, chain, residue count, bound ligand/cofactor.

Step 2 — Retrieve the Raw PDB Structure

- Download the selected structure in PDB format from RCSB PDB — this is the original/raw experimental model.
- Keep the original file unchanged; do not overwrite it. Use a clear naming system, e.g. 2ITU_original.pdb.

Expected Output:

- Original PDB file
- Screenshot of the selected structure
- Basic structural information

Step 3 — Inspect the Raw Structure

- Before refining, check: chain(s) and chain ID, residue count, missing residues, ligands, water molecules, alternate conformations, visible structural abnormalities.

Note: GalaxyRefine requires a single protein chain — ensure the submitted structure meets this.

Expected Output:

- Brief description of structural composition and key observations.

Step 4 — Pre-Refinement Validation

Validate the original structure using three tools before any refinement:

4.1 ERRAT

- Evaluate overall quality / unfavorable non-bonded interactions.
- Record: overall quality factor, ERRAT plot, key unfavorable regions.

4.2 PROCHECK

- Evaluate stereochemical quality via the Ramachandran plot.
- Record: most favored / additionally allowed / generously allowed / disallowed regions, % outliers.

4.3 Verify3D

- Evaluate compatibility between the 3D structure and its amino-acid sequence environment.
- Record: score/result, % residues passing criterion, overall interpretation.

Expected Output:

- Save results and screenshots from ERRAT, PROCHECK, and Verify3D.

Step 5 — Prepare the Pre-Validation Results

Compile initial validation results into one table and briefly explain what they indicate about the original structure.

Parameter	Raw Structure
ERRAT Quality Factor	
Ramachandran Favored (%)	
Ramachandran Allowed (%)	
Ramachandran Outliers (%)	
Verify3D Result	
Overall Structural Quality	

Step 6 — Identify Structural Quality Issues

- Look for: Ramachandran outliers, low-quality ERRAT regions, poor 3D–1D compatibility, unfavorable residues, other irregularities.

Note: Do not automatically label the structure “bad” — this step establishes the baseline quality before refinement.

Step 7 — Structure Refinement Using GalaxyRefine

- Submit the single-chain PDB structure to GalaxyRefine.
- Workflow: Original PDB → Upload → Refinement → Generated Models → Model Evaluation → Model Selection.
- GalaxyRefine may generate several models — select the most appropriate one based on reported quality indicators, not arbitrarily.

Expected Output:

- Save the selected refined structure as PDBID_refined.pdb
- Record relevant GalaxyRefine output information

Step 8 — Post-Refinement Validation

Run the same three validation procedures on the refined structure:

- ERRAT** — record the new quality factor and plot.
- PROCHECK** — record the new Ramachandran statistics.
- Verify3D** — record the new structural compatibility result.

Note: Use the same validation methodology as the original structure so the comparison stays meaningful.

Step 9 — Compile Post-Refinement Results

Parameter	Refined Structure
ERRAT Quality Factor	
Ramachandran Favored (%)	
Ramachandran Allowed (%)	
Ramachandran Outliers (%)	
Verify3D Result	
Overall Structural Quality	

Save all post-refinement screenshots.

Step 10 — Pre- vs Post-Refinement Comparison

The central analysis of the project — compare original and refined structures quantitatively.

Validation Parameter	Before	After	Change
ERRAT Quality Factor			
Favored Regions (%)			
Allowed Regions (%)			
Outliers (%)			
Verify3D Result			

Calculate the observed change where appropriate; state whether it is an improvement, no substantial change, or a deterioration.

Note: Refinement need not improve every parameter — mixed results are scientifically valid.

Step 11 — Structural Visualization

- Use PyMOL or ChimeraX to compare the original and refined models.
- Generate images of: (1) original structure, (2) refined structure, (3) superimposed original + refined, (4) important mutation/functional region, (5) ligand-binding region if applicable.

Expected Output:

- High-quality figures suitable for the final report.

Step 12 — Oncology-Based Structural Interpretation

- Discuss: the protein's role in cancer, importance of the selected mutation (if any), functional significance of the structure, relevant ligand/binding site, structural changes after refinement, and whether refinement affects interpretation of key structural regions.

Note: Keep the discussion evidence-based — do not claim biological effects that were not directly investigated.

Step 13 — Assess Suitability for Downstream Research

- Discuss whether the refined structure appears suitable for further computational studies: binding-site analysis, protein–ligand interaction analysis, molecular docking, virtual screening, structure-based drug discovery.

Note: Distinguish structural quality improvement from actual experimental confirmation of biological/drug-binding effects.

Step 14 — Final Scientific Conclusion

Central question: *Did computational refinement improve the overall structural quality of the selected oncology-related protein structure?*

Base the conclusion on combined evidence: ERRAT + PROCHECK + Verify3D + Structural Comparison. State clearly:

- What changed?
- Which parameters improved?
- Which parameters remained unchanged?
- Were any parameters worse after refinement?
- What does the overall result indicate?
- What are the limitations of the analysis?

Final Project Deliverables

Structural Files

- Original PDB structure
- Refined PDB structure

Validation Results

- Pre-refinement: ERRAT, PROCHECK, Verify3D
- Post-refinement: ERRAT, PROCHECK, Verify3D

Comparative Analysis

- Pre vs Post results table
- Interpretation of changes
- Original vs refined structural visualization

Final Report Structure

1. Title · 2. Abstract · 3. Introduction · 4. Objective · 5. Protein/PDB Information · 6. Methodology · 7. Pre-Refinement Validation · 8. GalaxyRefine Refinement · 9. Post-Refinement Validation · 10. Pre vs Post Comparison · 11. Structural Visualization · 12. Oncology-Based Interpretation · 13. Discussion · 14. Limitations · 15. Conclusion · 16. References.

Final Project Workflow

Oncology Protein Selection → PDB Retrieval → Raw Structure Inspection → ERRAT + PROCHECK + Verify3D → Pre-Refinement Quality Assessment → GalaxyRefine → Refined Structure Selection → ERRAT + PROCHECK + Verify3D → Pre vs Post Comparison → 3D Structural Visualization → Oncology-Based Interpretation → Scientific Conclusion → Final Project Report